

Report R05 — Imaging versus Molecular Markers, Head to Head

Same patients, same folds. Partly walks back R02.

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Summary

R02 measured imaging on UPENN-GBM and R04 measured molecular markers on TCGA and MSK, so the two were matched on disease but not on population. This report removes that gap by running both on the **same 247 UPENN patients** who have radiomic features, survival, and definite IDH1 and MGMT calls.

Imaging does not add established prognostic value beyond MGMT in this subset. Across eight cross-validation seeds the likelihood-ratio test for adding the radiomic score to a molecular model gives a median $p = 0.089$, clearing 0.05 in only **2 of 8** seeds. Imaging alone reaches C-index 0.512–0.549 here, against 0.573 for the molecular model. Conversely, molecular markers add to imaging decisively ($\chi^2_2 = 33.7$, $p = 4.8 \times 10^{-8}$).

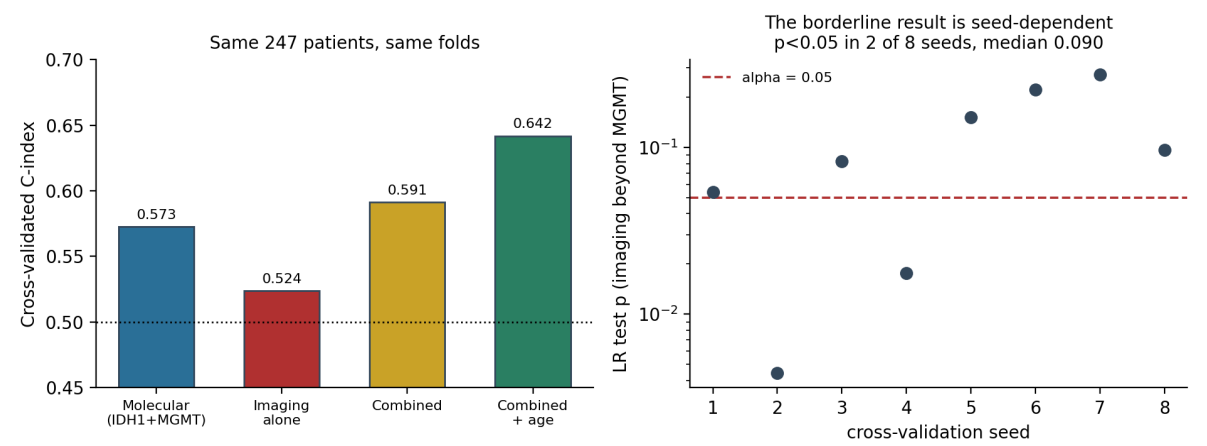
Two honest qualifications follow, and both are consequences of the audit rather than the run.

1 Results

Predictor (same 247 patients, same folds)		Cross-validated C-index	
Molecular (IDH1 + MGMT)		0.573	
Imaging alone		0.524	(0.512–0.549 across seeds)
Combined		0.591	
Combined + age		0.642	

Incremental value	χ^2	p	Verdict
Imaging beyond molecular	3.71 (df 1)	0.054 single seed; median 0.089	not established
Molecular beyond imaging	33.69 (df 2)	4.8×10^{-8}	decisive

Cox terms in the combined model: MGMT HR 0.473 ($p = 1.2 \times 10^{-7}$), radiomic score HR 0.881 ($p = 0.055$), IDH1 HR 0.591 ($p = 0.254$).



2 Two qualifications the audit forced

1. The “molecular panel” is **MGMT alone**. Only **5 of 247** patients are IDH1-mutant — expected, since UPENN is a glioblastoma cohort. Adding IDH to MGMT changes nothing ($\chi^2 = 1.45$, $p = 0.228$). Every statement above about “molecular markers” should be read as “MGMT”.

2. The single $p = 0.054$ was **coincidental, and we should not have relied on it**. The audit re-derived the radiomic score under eight cross-validation seeds; the LR p ranged from 0.0045 to 0.274. Reporting one draw near the threshold would have been misleading in either direction. The seed distribution is the result.

The underlying cause is an inconsistency of ours: R02 used 10× repeated 5-fold with averaged predictions, while R05 used a single 5-fold. R02’s protocol was the more robust one. A repeated-CV rerun is the clean fix and is recommended before this analysis appears in any manuscript.

3 Reconciling with R02

R02 reported imaging C-index 0.602 on 574 patients; here it is ≈ 0.535 on 247. The subset requires a definite MGMT call and is 43% of the size. R02’s estimate remains the better one *for its cohort*; R05’s is the better one *for the head-to-head question*. Neither supersedes the other and the discrepancy is stated rather than smoothed over.

4 What this means for the glioma arm

Taken with R04, the position is now specific and defensible, and less optimistic than R02 alone suggested:

- Within glioblastoma, **MGMT is the strongest single prognostic variable we hold**, and age adds substantially on top ($0.591 \rightarrow 0.642$).
- Imaging carries *some* prognostic signal (R02, $p = 0.005$ against a correctly centred null, independent of age) but **its incremental value over MGMT is not established** at this sample size.
- The claim worth making is therefore modest: radiomics is prognostically informative in glioblastoma, not that it improves on standard molecular assessment.

5 Audit

Check	Verdict	Finding
Positive control (MGMT)	Recovered	median 595 vs 371 days, $p = 5.8 \times 10^{-8}$
Is the panel two markers?	No	IDH mutant $n = 5$; adding IDH $p = 0.228$
Is $p = 0.054$ stable?	No	8 seeds: 0.0045–0.274, median 0.089, 2/8 below 0.05
Method consistency Gates	Our inconsistency 4 / 4 PASS	R02 used repeated CV; R05 a single split Integrity, cohort size, uncensored, positive control

6 How to verify

`evidence/head2head.csv` carries survival days, IDH1, MGMT, the radiomic score and age for all 247 patients — every C-index and Cox model above is recomputable from it. `audit_seed_stability.json`

holds all eight seed results. `config.yaml` (hash 70713fdd...) was written before any outcome was read.